

TECHNICAL PROGRESS REPORT

Source-Linked Compression-Set Data for Silicone Formulations

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Executive summary

This report summarizes the work completed to build a source-linked database of silicone formulations and compression-set measurements from patents and journal articles. It describes the extraction pipeline, the frozen dataset it currently produces, and the predictive baseline that dataset supports. The work targets a practical question for seal design. Given a documented recipe (base polysiloxane, filler, crosslinker, catalyst or initiator, additives) and a test protocol, can compression set be predicted without running the test?

The central result is a document funnel: the published record is large but almost entirely qualitative, and the usable quantitative record becomes smaller at every screening step. We began with 2,925 downloaded PDFs. Of these, 1,930 yield at least one extracted property. Only 183 report any compression-set value at all, and once a value is required to sit next to a stated composition and a complete test protocol (temperature, time, deflection), the population shrinks to 85 documents. The clean release `cs_v14` holds 1,054 measurements over 605 formulations from 67 documents (65 patents, 2 journal articles). Every value in it was read from a page image by a locally served vision-language model, resolved to a canonical substance and role by a reviewed language-model prompt, checked against a verbatim source quote, and rebuilt from inputs rather than patched in place. A two-reader audit of a stratified sample of released rows against their source pages found the released compression-set value correct in 96.8% of checkable cases (Wilson 95% interval 94.7–98.1%), with temperature and deflection correct in every checked case.

The release also separates the formulation classes that matter for interpreting the results. We call a formulation a *composite* when its reviewed role labels include a dispersed filler (reinforcing, functional, extending, or lightweight filler), and a *foam* when its labels include a blowing agent. These labels can overlap: 525 formulations are composites and 115 are foams, including 110 formulations that are both. Thus the release contains 415 composite-only formulations, 5 foam-only formulations, and 75 formulations in neither class. This classification is used descriptively and does not replace the source’s own product terminology.

On this release, test conditions alone (deflection, time, temperature) give a leave-one-family-out

R^2 of 0.289 with a mean absolute error of 17.5 percentage points. Adding any of four chemistry representations (stated viscosity, ingredient-text embeddings, a blend of text and parsed descriptors, ingredient categories) changes that score by between +0.023 and +0.090, and every one of those intervals includes zero. At 67 documents the smallest gain a paired test could distinguish from zero lies between 0.10 and 0.23 R^2 , larger than any chemistry effect observed. The corpus therefore supports a conditions-only baseline and a documented dataset, and it does not yet support any claim that composition features improve transfer to unseen patent families. Random-row cross-validation on the same rows reaches R^2 of 0.737, which measures interpolation within known documents and is reported only as a diagnostic.

Two items are requested from collaborators. First, a small set of formulations with trusted chemistry and test conditions that could serve as a fixed validation set. Second, a statement of the prediction error that would change a design decision, so that the corpus size needed to prove a useful model can be estimated rather than assumed.

Table 1: Headline numbers. All counts refer to release cs_v14 (frozen 2026-09-13) unless stated otherwise.

Downloaded PDFs	2,925
Documents with any compression-set value	183
Documents in the clean release	67 (65 patents, 2 journal)
Formulations released	605
Compression-set measurements released	1,054
Composite formulations / documents	525 / 60
Foam formulations / documents	115 / 17
Ingredient rows released / distinct ingredient strings	4,898 / 945
Median compression set (interquartile range)	25% (13–51%)
Distinct test protocols (temperature, time, deflection)	60
Released value correct on re-read of source page	96.8% [94.7, 98.1]
Leave-one-family-out R^2 , conditions only (ridge)	0.289
Best chemistry arm, leave-one-family-out R^2	0.280 (text embeddings)
Smallest provable gain at 67 documents	0.10–0.23 R^2

1 Goals and scope

1.1 Why compression set

Silicone elastomers combine thermal stability, chemical inertness, and elastic recovery, which is why they dominate seals that must hold a load for years [1, 2]. Compression set is the fraction of an imposed deflection that a specimen fails to recover after a stated time at a stated temperature [3]. It is the single property most directly tied to seal life, and it is also the property whose value depends most on how the test was run. A button held 22 h at 175 °C under 25% deflection and a button held 70 h at 150 °C are different experiments, and a value reported without those three numbers cannot be compared to anything. The data model in this project therefore treats the test protocol as part of the measurement rather than as metadata.

The property depends on network chemistry as much as on protocol. Residual set arises from chain slippage at filler surfaces, from post-cure crosslinking under the test load, and from scission

and re-formation of siloxane bonds at temperature, so the base polymer’s end groups and vinyl content, the crosslinker’s Si–H content, the filler’s surface area and treatment, and the choice of an addition, condensation, or peroxide cure all enter [1, 4, 5]. A useful dataset must carry that chemistry in a form a model can read.

The broader difficulty is well documented. Polymer formulations resist data-driven modeling because the design space is large and much of the practical knowledge is recorded qualitatively [6–8], and the experimental record that could anchor models is dispersed across patents and articles with incompatible layouts, trade-name opacity, and reprinted data [9, 10]. Curated polymer databases such as PoLyInfo [11] cover homopolymer properties and not formulated elastomers, and the polymer language models built on them [12–14] take a repeat-unit string as input, which a filled, catalyzed, post-cured network does not have. Text-mined materials databases have been built for inorganic synthesis recipes, solar-cell devices, and magnetic transition temperatures [15–18], first with rule-based toolkits [19] and more recently with large language models prompted to return structured records [20–24]. Those efforts read text. Formulation tables in patents are laid out visually, span pages, and rotate, which is why this project reads rendered pages with a vision-language model instead.

1.2 The state of the published record

Figure 1 surveys the record from which the dataset is built. Panel (a) is the main document funnel and should be read from top to bottom: the count decreases at each requirement. The starting set contains 2,925 PDFs; 1,930 yield at least one extracted property of any kind; 183 report a compression-set value (3,961 raw rows); 161 attach such a value to a stated composition; 85 also state the full test protocol; and 67 survive the remaining checks into the clean release. Each step in that funnel is a source that describes a material qualitatively without giving the numbers needed to use it quantitatively.

Panel (b) shows where the numbers go missing. Of 3,862 compression-set rows that are neither prophetic nor out of range, 92% carry a value in the row and 80% sit next to a composition, but the row itself states a test temperature for only 46%, a time for 50%, and a deflection for 14%. All fields appear together in the row for 7% of measurements. The protocol is usually written once, in a methods paragraph pages from the results table, and sometimes only as the name of a standard. Reading the whole document and resolving those statements back onto each row raises deflection coverage to 81%, temperature to 72%, and full specification to 50%. Panel (c) reports the formulation classes. A composite is counted when a formulation contains a reviewed dispersed-filler role; a foam is counted when it contains a reviewed blowing-agent role. The classes overlap, so the composite-plus-foam bar is not added a second time to obtain the total of 605 formulations. The pipeline’s job is to turn the surviving record into rows in which chemistry, amount, protocol, and value are all present and all traceable to a page.

1.3 What this report covers

Sections 2 to 5 describe the silicone compression-set effort, which is the primary deliverable. Section 6 records verification cases that changed the pipeline’s design. Section 7 summarizes a companion benchmark on a 1968 organosilicon reference volume that was used to develop and test the table-reading tools in a setting where every answer can be checked. Section 8 lists the experiments planned next and the input requested from collaborators. Appendices give the data dictionary of the release tables and the commands that regenerate every number and figure in this report.

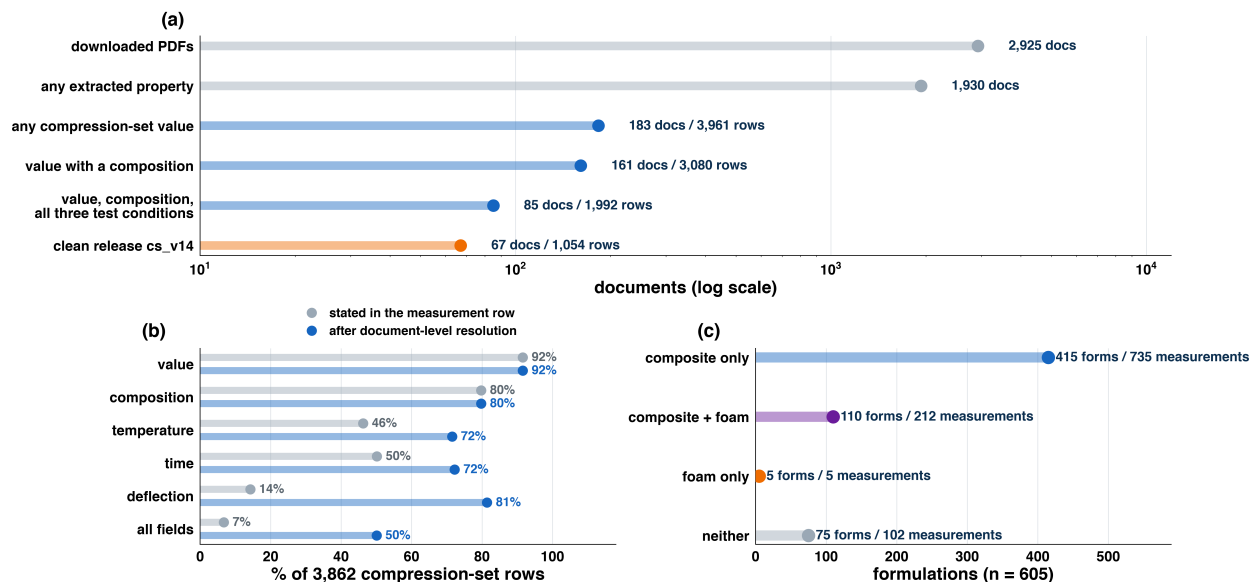


Figure 1: The document funnel and formulation classes. (a) Documents remaining at each stage from the downloaded PDFs to the clean release; the count decreases as composition, protocol, and release-quality requirements are imposed, and row counts are given where the stage is defined on measurements. (b) Share of 3,862 non-prophetic, in-range compression-set rows for which each field is stated in the measurement row itself (grey) and after document-level resolution of the methods text (blue). (c) Formulations classified as composite, foam, both, or neither. Composite means a dispersed-filler role; foam means a blowing-agent role.

2 What was accomplished: the extraction pipeline

Figure 2 lays out the pipeline in three lanes. The first finds and screens documents, the second reads them into candidate records, and the third rebuilds a database from those records, freezes a release, and evaluates models on it. Two conventions run through all three. Every document carries a permanent opaque identifier assigned at first sighting, so richer metadata learned later never renames it. And every data file is regenerated by rerunning the stage that produces it. When a value is wrong, the fix goes into the stage or into a reviewed correction file, and the stage reruns; no released CSV is edited by hand.

2.1 Discovery and acquisition

Documents come from five sources. Patents are harvested from the USPTO Open Data Portal and from the European Patent Office’s Open Patent Services (EP and WO publications); journal articles are located through OpenAlex, Semantic Scholar, and the OSTI grey-literature index. A single declarative specification of material synonyms, form terms, and test standards renders the query for every source, so a synonym added once reaches every source’s next harvest. The specification deliberately omits the property from the query. Conjoining “compression set” with the material terms collapsed USPTO recall almost to zero, because patents state the property in an example table rather than in the title or abstract, and in a screening pilot on OSTI grey literature only 2 of 76 candidates showed compression-set evidence in the abstract against 8 of 14 when the full text was fetched. The property judgment is deferred to a full-text screen.

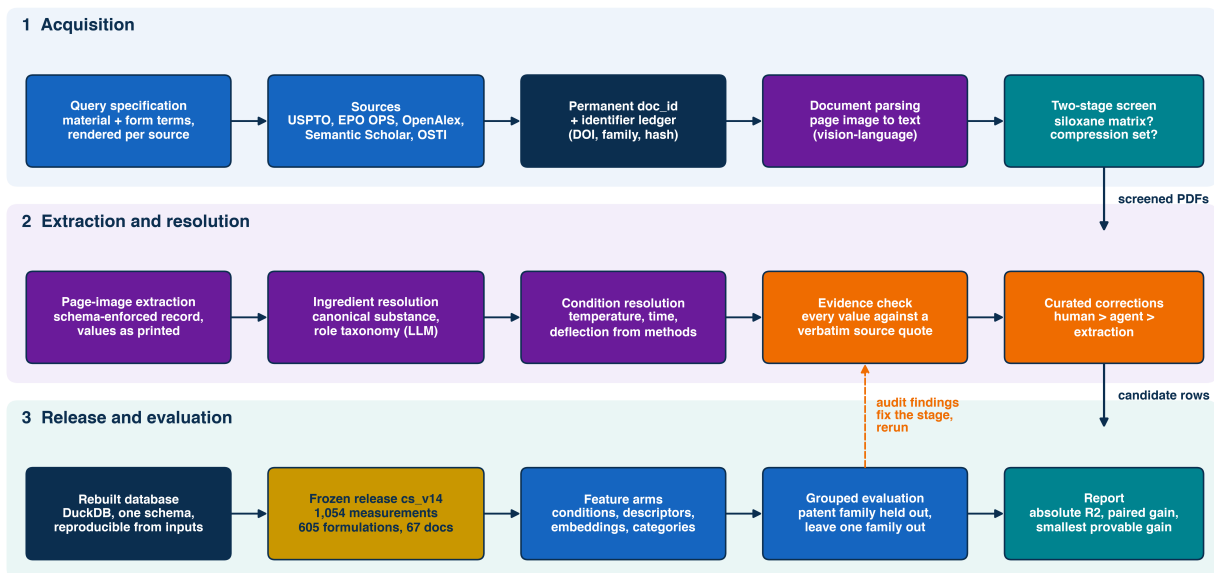


Figure 2: From source discovery to grouped evaluation. Lane 1 renders one query specification against every source, assigns a permanent document identity, parses each PDF to text once, and applies a two-stage language-model screen. Lane 2 reads page images into schema-enforced records, resolves ingredient identity, role, and test conditions with reviewed prompts, checks each value against a verbatim quote, and applies curated corrections. Lane 3 rebuilds the database, freezes a release, builds feature arms, and evaluates with patent families held out. Audit findings from lane 3 return to lane 2 as fixes to a stage, never as edits to a release.

Duplicates are removed by identity rather than by text. A DOCDB patent family identifier, a DOI, a content hash, and provider identifiers are appended to an identity ledger under the permanent document id, and a curated subset of high-confidence identifiers merges duplicates automatically. This is also the mechanism that recognizes an EP or WO family member as a reprint of a US document already in hand and keeps only the fullest one. Family identity matters again at evaluation time, because rows from two members of one family are near-duplicates and must never be split across training and test folds (Section 5.2).

Every downloaded PDF is transcribed to structured text by a vision-language parser before any question is asked of it. Screening is then a text-classification problem asked of a local language model in two stages. The first asks whether the matrix polymer is a polysiloxane; only documents that pass are asked whether they report quantitative compression-set measurements. An audit mode reruns the second stage on a sample of first-stage rejections to measure the recall the cascade costs. No pattern matching enters either judgment; a keyword search is used only to select which passages of a long document are shown to the model. We began with 2,925 downloaded PDFs: 314 journal articles and 2,611 patents. The silicone-matrix screen retains 2,409 documents. Of those, 165 (54 journal articles and 111 patents) also report quantitative compression-set measurements.

2.2 Page-image extraction

Each PDF page is rendered to an image and passed to a vision-language model (Qwen3-VL-32B-Instruct [25, 26], served locally with vLLM [27] on an NVIDIA H200) with a reviewed prompt and a JSON schema. The model returns, for each formulation on the page, every ingredient’s verbatim

name, role text, amount, and unit, the processing conditions, and the measured properties with their printed conditions. Values are captured as printed. No unit conversion, synonym substitution, or inference from a trade name happens at this stage, so the record can later be checked cell by cell against the page it came from. The record also flags prophetic examples, keeps the page number and table label, and records the model’s own confidence. Reading the image rather than extracted text matters for patents, where a composition table often spans pages, rotates, or repeats a header, and where the text layer of a scanned document is unreliable; the same argument motivated OCR-free document understanding models [28, 29].

An independent text-based path (a commercial document parser feeding the same schema through a text-only model) runs alongside the image path on every document. Neither dominates. Comparing the two per document catches merged or flattened tables that either one alone would pass silently, and the disagreement census is one of the pipeline’s standing quality checks. The release described here uses the image path for every document.

2.3 Resolution of ingredients, roles, and conditions

A printed ingredient string becomes usable through three resolutions, each performed by a local language model under a prompt that was reviewed before it ran, each cached with the model identifier and prompt hash, and each applied only after its output has been checked against source quotes.

Identity resolution maps the verbatim string to a canonical substance and, where the document defines it, to a CAS number or structure. Trade codes and anonymous labels are resolved from the document’s own glossary sentences (“component (A) is a dimethylvinylsiloxyl-terminated polydimethylsiloxane having a viscosity of 10,000 mPa·s”), and the descriptive paragraph itself is kept as the canonical text because it carries the chemistry a model can use. Role resolution assigns each ingredient one or more labels from a reviewed taxonomy of 41 leaves under 16 parents (base polymer, crosslinker, cure catalyst, cure initiator, cure inhibitor, reinforcing filler, extending filler, filler treating agent, and so on). The distinction between catalyst and initiator is chemical rather than nominal. Platinum, tin, and titanium compounds in addition and condensation cures are regenerated and are catalysts; peroxides and azo compounds are consumed and are initiators, even when the source calls its peroxide a “curing catalyst”. No regular expression is used for any chemical classification; the taxonomy and a reviewed prompt are the only classifier.

Condition resolution recovers the test protocol from wherever the document states it. A string-level pass reads the condition text printed beside a value; a document-level pass reads the methods text and the standard named, distinguishes a test hold from a pre-aging exposure, and attaches temperature, time, deflection, and medium to each measurement with a verbatim quote as evidence. Each resolved field carries its source (row text, methods statement, human verification, or explicit assumption) so a downstream user can filter on provenance.

2.4 Verification and correction

Agent output is a candidate until it has been checked against the source and promoted, and a candidate is never written into a curated file. Verification takes three forms. Every resolved value must be accompanied by a verbatim quote from its own document, and quotes that cannot be located on the cited page fail the check. A sample of released rows is re-read from the page images by two independent readers and compared cell by cell (Section 4.3). Compositions that collapse to

identical ingredient sets with different measured values are treated as detectors of dropped variables or parser failures and are sent back for repair rather than tolerated as a dataset ceiling.

Corrections are centralized in a set of schema-validated correction files (curated formulation components, curated conditions, glossary overrides, amount-basis overrides, translations). A human correction beats an agent correction, which beats the raw extraction. Each correction row carries its source, reviewer, and date. Documents in which silicone is an additive or crosslinker in a non-silicone matrix (polyurethane, thermoplastic vulcanizate, styrenic block copolymer) are out of scope and are quarantined through a script that archives rather than deletes.

2.5 Rebuild and freeze

Normalized records, the corpus index, the resolution caches, and the correction files are loaded into a single DuckDB store by one build script. The build is a fixed point of its inputs, verified empirically by two consecutive rebuilds that produced bit-identical content for every table except the one that records the build’s own git commit. A release is a frozen snapshot of that store passed through an admission rule (usable, non-prophetic, in physical range, numeric target, stated composition, all three test conditions present, recipe complete enough to model) together with a manifest that pins the cohort file, the database build stamp, an archive of the machine caches, and the SHA-256 of every correction file. The modeling code reads only the frozen snapshot, never the live database.

3 Data model and conventions

3.1 Formulation composition

A formulation is a list of ingredient rows. Each row carries the verbatim name, the resolved canonical substance, the role from the taxonomy, the amount with its basis (parts per hundred of base polymer, weight percent of the final formulation, grams, ppm), and any characterization the source states (viscosity, molecular weight, vinyl or Si-H content, BET surface area, particle size, surface treatment). The amount basis is recorded per row and summarized per formulation, because a recipe that mixes bases cannot be converted to fractions without an assumption the source did not make. Patent examples that describe multi-part systems (Part A and Part B, or liquids I, II, and III) are split into per-part formulations, since a half-recipe that cannot cure on its own would otherwise be modeled as if it were a complete material.

3.2 The compression-set measurement

Each measurement is one row. It carries the value in percent, the test temperature, the test time, the applied deflection, the medium (air, a named oil, water, or humid air), the specimen geometry where stated (disc or O-ring, thickness, diameter), the test standard where named, and any post-cure stated for that specimen. An aging series is several rows, one per protocol, and is never collapsed to one. Where the source distinguishes a pre-aging exposure from the test hold, both are kept as separate fields; the two are different experiments, and the pipeline is built so that a mirror of one into the other cannot manufacture agreement.

3.3 Not stated versus not applicable

An empty cell in the release means the source does not state the value. A field that does not apply to a row is never left empty silently. The clean release tables carry only fields that apply to every row; fields that apply to some rows (post-cure, pre-aging) live in the provenance tables with an explicit flag. Provenance is tracked at the field level. A measurement’s temperature can come from the row text, from a methods statement, from a human verification, or from an explicit assumption, and the release records which. A user who wants only human-verified protocols can filter for them.

3.4 Scope

The corpus keeps only materials whose matrix is a polysiloxane. Blends in which a silicone is the minor phase, a crosslinker, or a processing aid are excluded, as are epoxy, polyurethane, and thermoplastic-vulcanizate systems that merely contain a siloxane. The screen is applied by a language model reading the document, and its rejections are audited. One journal article on a DGEBA epoxy modified with a polydimethylsiloxane was retrieved by the material query, passed the keyword filter, and was excluded only by the matrix screen; it is the canonical example of why the screen reads the document rather than the abstract.

4 Significant result: dataset release cs_v14

4.1 Contents

Release cs_v14 was frozen on 13 September 2026 from a database build of the same day. It holds 1,054 compression-set measurements over 605 formulations from 67 documents (65 patents and 2 journal articles published between 1974 and 2026), 4,898 ingredient rows, and 945 distinct ingredient strings, every one of them in English and judged either chemically specific or a typed generic (“fumed silica” passes; “component E2” does not). No formulation and no measurement was held back at export; earlier releases had held back rows whose ingredients could not be named, and the glossary and translation work that closed that gap is part of what separates cs_v14 from its predecessors.

Figure 3 summarizes the release. The compression-set values have median 25% and interquartile range 13–51%, with a long tail to 100% that comes mostly from aged specimens and comparative examples (panel a). Measurements are unevenly distributed across documents, with a median of 8 per document and a maximum of 93 (panel b); this concentration is the reason evaluation must hold out whole documents. Amount bases are mixed (panel c): 447 formulations are in parts per hundred, 124 in weight percent, 11 in grams, and 18 mix bases within one recipe. A base polymer is named in 98% of formulations, a crosslinker in 73%, a cure catalyst in 68%, and a cure initiator in 30% (panel d); the two cure agents are close to mutually exclusive, as the chemistry requires. The 60 distinct test protocols cluster on the standard industrial holds of 22 h at 175 °C (124 measurements) and 70 h at 150 °C (28 measurements) but extend from room temperature to above 200 °C and from hours to thousands of hours (panel e). Panel (f) records what the sources state. Temperature, time, and deflection are present on every released row by construction. Medium is filled for every row (air unless the source names another medium), specimen shape is stated for 60%, specimen thickness for 51%, and a test standard for 49%. Those gaps are not filled; they are the record.

4.2 Which formulations are composites and which are foams?

The formulation classes are reported explicitly because they change the interpretation of both chemistry and compression-set comparisons. The labels below are derived from the reviewed ingredient-role taxonomy, not from a keyword applied to the document title. A composite requires at least one dispersed filler role: reinforcing, functional, extending, or lightweight filler. A foam requires a blowing-agent role. A formulation can satisfy both conditions, as it does when a filled silicone is expanded into a cellular material.

Table 2: Operational material classes in release cs_v14. The composite and foam labels can overlap; the four rows below partition the formulations.

Formulation class	Formulations	Compression-set measurements
Composite only	415	735
Composite and foam	110	212
Foam only	5	5
Neither composite nor foam	75	102
Total	605	1,054

In aggregate, 525 formulations from 60 documents are composites and 115 formulations from 17 documents are foams. The overlap is substantive rather than an accounting error: the 110 formulations in both classes contribute to both totals.

4.3 Accuracy of the release

Sixty released measurements were sampled, stratified by extraction path, script, and source type, and each was traced to its source page. A reader transcribed the page image, compared each parser’s cell output against the transcription, and checked the released row; seven pages received a second, blind read. Of 437 checkable compression-set values the released value was correct for 423 (96.8%, Wilson 95% interval 94.7–98.1%). Temperature was correct in every case, deflection in every case, and time in 99.7%. Ingredient amounts were correct in 97.0% of 332 checkable cells [94.5, 98.4] and the amount basis in every case. Inter-reader agreement on the seven double-read pages was 81%. At the parser level, the image-based path reproduced 85.6% of table cells [84.4, 86.7] and the text-based path 75.8% [74.1, 77.5]; the difference is dominated by merged or flattened tables and by rows shifted across columns.

The audit produced a short list of confirmed errors that remain in cs_v14 and will be fixed in the next freeze through the correction files rather than by editing the release. Nine formulations in one US patent carry values from a neighbouring table under the wrong condition (released 14–55% against printed 71–98%). One amount in another US patent is the neighbouring row’s value (1.54 against printed 1.15). One inhibitor amount was read as 0.4 against a printed 0.04, and one time as 1,008 h against a printed 1,000 h. Each is listed with its source page in the audit summary so that a user of cs_v14 can exclude the affected rows.

4.4 Release history

Releases are immutable and named. The first pinned cohort after the adversarial audit of machine-reconstructed tables held 796 rows from 45 documents. Release cs_v11 held 873 rows from 65

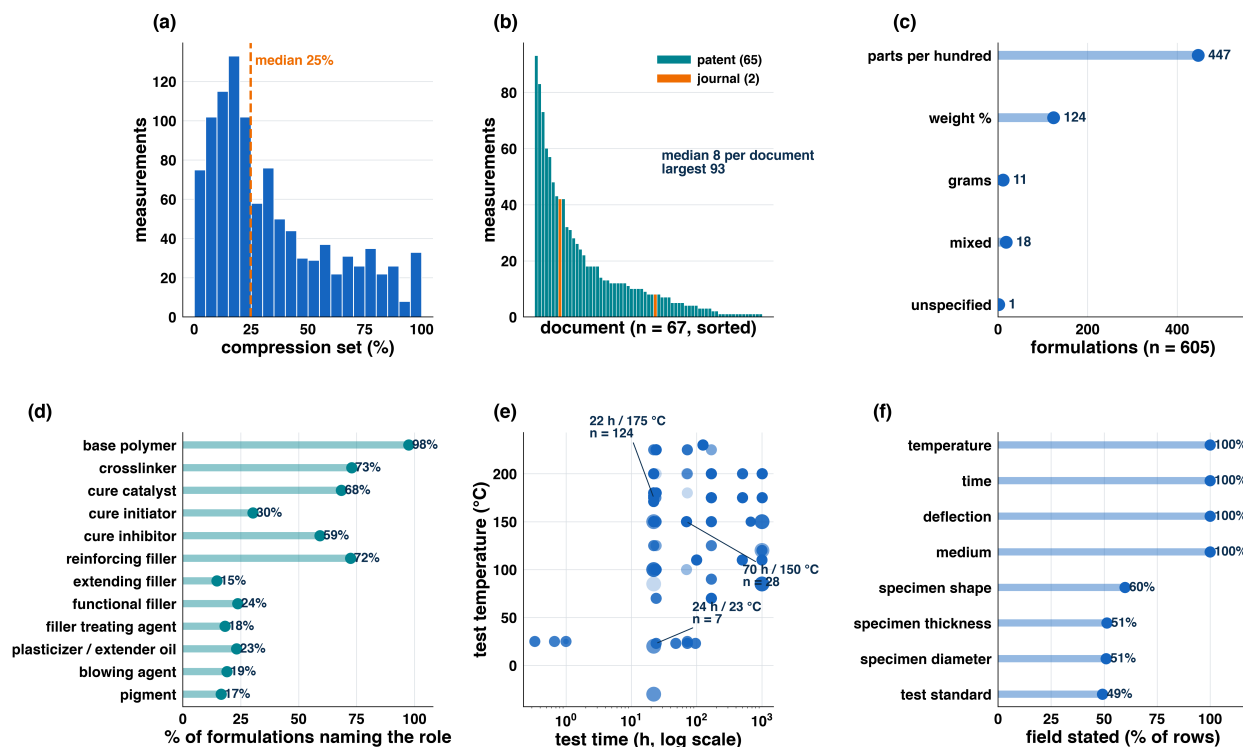


Figure 3: Contents of release cs_v14. (a) Distribution of the 1,054 compression-set values with the median marked. (b) Measurements per document, sorted, with journal articles in orange. (c) Amount basis per formulation; a recipe is “mixed” only when two weighable bases appear together, and ppm-level entries do not count. (d) Share of formulations naming each ingredient role. (e) Test protocols as temperature against time, with marker area proportional to deflection and the three most common protocols labeled. (f) Share of measurements for which the source states each field; empty cells in the release mean the source is silent.

documents; cs_v12 (2 September 2026) held 856 rows, 447 formulations, and 58 documents after five patent-family reprint pairs were quarantined; cs_v13 (11 September) held 1,043 rows, 593 formulations, and 67 documents; cs_v14 (13 September) holds 1,054, 605, and 67. Between cs_v13 and cs_v14 two documents were quarantined (one out of scope, one with a defective extraction) and one was added. Numbers reported at earlier meetings belong to the release they were computed on and are not comparable across releases without that qualification.

5 Significant result: a predictive baseline

5.1 Target and feature arms

The target is the compression-set value of each row. The conditions floor uses four columns derived from the three required conditions: deflection, $T/1000$ in kelvin, $\log_{10} t$ in hours, and their product $T \log_{10} t / 1000$. The last three span the Larson–Miller family of time-temperature parameters, so the collapse constant is fitted rather than assumed. Every other arm appends one block of chemistry to those four columns, and every arm is evaluated on identical rows and folds, so each score is a paired change against the same floor.

Four chemistry blocks are compared. *Stated viscosity* is the base polymer’s viscosity where the

source gives it, with a flag for “not stated”; it is the one numeric descriptor with enough coverage to test alone. *Ingredient-text embeddings* encode each ingredient’s resolved description with a materials-science text encoder (MatSciBERT [30]), pool the vectors within a formulation weighted by amount, and reduce them inside each training fold. *Text and descriptors* is a prediction-level blend, with fixed equal weights, of the embedding arm and an arm of parsed physical descriptors (viscosity, molecular weight, vinyl and Si-H content, filler surface area, particle size, surface treatment) that records only values the source states. *Ingredient categories* is the presence and amount-weighted share of each taxonomy role in the formulation. Ridge regression carries the floor; gradient-boosted trees [31, 32], with preprocessing in scikit-learn [33], carry the chemistry arms, and the tree estimator’s own conditions-only score is reported so that the paired gain is always measured against the same estimator.

5.2 Evaluation protocol

Rows from one patent family are near-duplicates, and random splitting of such data leaks the identity of the source into the test fold [34–36]. Every evaluation therefore groups by DOCDB family, with the document as the group where no family tag exists; the 67 documents form 66 groups. Three regimes are reported and named consistently. *Grouped five-fold* holds out whole families and averages over 30 fold-assignment repeats; the reported spread is fold noise, not an effect size. *Leave one family out* holds out each group in turn and pools the predictions; it is the regime used for the paired comparison. *Random rows* splits individual measurements and is an in-distribution diagnostic only, since siblings from one table sit on both sides of the split. The distinction between interpolation within known sources and extrapolation to new ones is the same one drawn for materials discovery benchmarks [37–39]. The gain of each chemistry arm is its leave-one-family-out R^2 minus the same estimator’s conditions floor. Its 95% interval comes from recomputing the gain while leaving out one document family at a time, and the *smallest provable gain* is the gain that a paired test could distinguish from zero at this corpus size. Scaling, imputation, and dimensionality reduction are fitted inside each training fold.

5.3 Results

Table 3 and Figure 4 give the ladder. The conditions floor reaches a leave-one-family-out R^2 of 0.289 (MAE 17.5 points) and a grouped R^2 of 0.258. The text-embedding arm reaches 0.280 and the blend 0.278, each within a few hundredths of the floor; stated viscosity and ingredient categories fall below it. Against the tree estimator’s own floor (0.190), the paired gains are +0.023 for viscosity, +0.090 for text, +0.088 for the blend, and +0.037 for categories, with 95% intervals of [-0.074, +0.254] for the text arm and [-0.059, +0.234] for the blend. Every interval includes zero. The smallest provable gain at 67 documents is 0.10 for the viscosity arm and between 0.21 and 0.23 for the higher-dimensional arms, which is larger than any gain observed.

Random-row scores on the same rows are 0.401 for conditions alone and 0.737 for categories. The gap between those numbers and the grouped scores is the size of the document fingerprint. A model that has seen a patent’s other examples can predict the next one; a model that has not seen the patent gains little from knowing its chemistry, at least in the representations tried and at this corpus size.

Table 3: The ladder on release cs_v14 (1,054 measurements, 67 documents, 66 groups). Grouped R^2 is the mean over 30 fold-assignment repeats of five-fold cross-validation with patent families held out. Random R^2 splits individual rows and measures interpolation within known documents. Leave-one-family-out (LOFO) R^2 and MAE are pooled over held-out families. The gain is LOFO R^2 minus the same estimator’s conditions floor; its interval comes from leaving out one document family at a time; the last column is the smallest gain a paired test could distinguish from zero at this corpus size.

Estimator	Arm	Grouped R^2	Random R^2	LOFO R^2	MAE	Gain [95% interval]	Provable
ridge	conditions	0.258	0.401	0.289	17.5	floor	—
xgboost	conditions	0.145	—	0.190	—	floor	—
xgboost	+ stated viscosity	0.157	0.548	0.213	18.1	+0.023 [-0.046, +0.092]	0.10
xgboost	+ ingredient text	0.199	0.690	0.280	17.2	+0.090 [-0.074, +0.254]	0.23
xgboost	+ text and descriptors	0.234	0.669	0.278	17.5	+0.088 [-0.059, +0.234]	0.21
xgboost	+ ingredient categories	0.209	0.737	0.228	17.5	+0.037 [-0.115, +0.189]	0.22

5.4 What the result does and does not say

The result bounds what the present corpus can support. It does not say that composition is irrelevant to compression set, which would contradict both the chemistry and the within-document variation visible in the data. It says that at 67 source families, with chemistry represented as stated viscosity, text embeddings, parsed descriptors, or taxonomy categories, the transfer gain from chemistry is smaller than the gain the corpus could prove. Four independent increases in corpus size over the project’s history (a doubling of documents, an EPO harvest, a 20% increase in complete recipe-property pairs, and the present clean release) each left the grouped score near the floor, so document count alone is not the lever. A rough scaling of the provable gain suggests that proving a gain of 0.05 would need on the order of 130 well-documented source families. The alternative lever is a fixed, trusted validation set on which representation choices can be compared without the noise of unseen-document transfer, which is the request made in Section 8. Text embeddings of ingredient descriptions were tried because unsupervised embeddings have been shown to carry materials knowledge from the literature [40], and because recent polymer models pool language-model embeddings of constituent descriptions in the same way [41]; three encoders (a general sentence encoder [42], MatSciBERT, and a large general-purpose embedding model) were compared under the same paired harness in earlier releases and MatSciBERT was retained.

One caveat applies to the text arm as it was run. A check made while preparing this report found that the frozen embedding block carries a zero vector for 183 of the 605 formulations and only 124 distinct vectors in all, because the embedding cache had been built on an earlier generation of ingredient strings and cs_v14 renamed most of them (89 of 945 strings had a cached vector when the check was made on 16 September 2026, before the cache was refreshed). The text arm in Table 3 therefore saw no chemistry at all for three formulations in ten and the same vector for many of the rest, and its score is a lower bound on what the representation can do. The cache has been refreshed for the cs_v14 strings and the ladder will be rerun; the conditions floor and the viscosity and category arms are unaffected.

5.5 The ingredient descriptions as the encoder sees them

Figure 5 projects the 945 distinct ingredient descriptions in the release, embedded with the same MatSciBERT encoder the text arm uses, to two dimensions with UMAP. Colored by taxonomy role

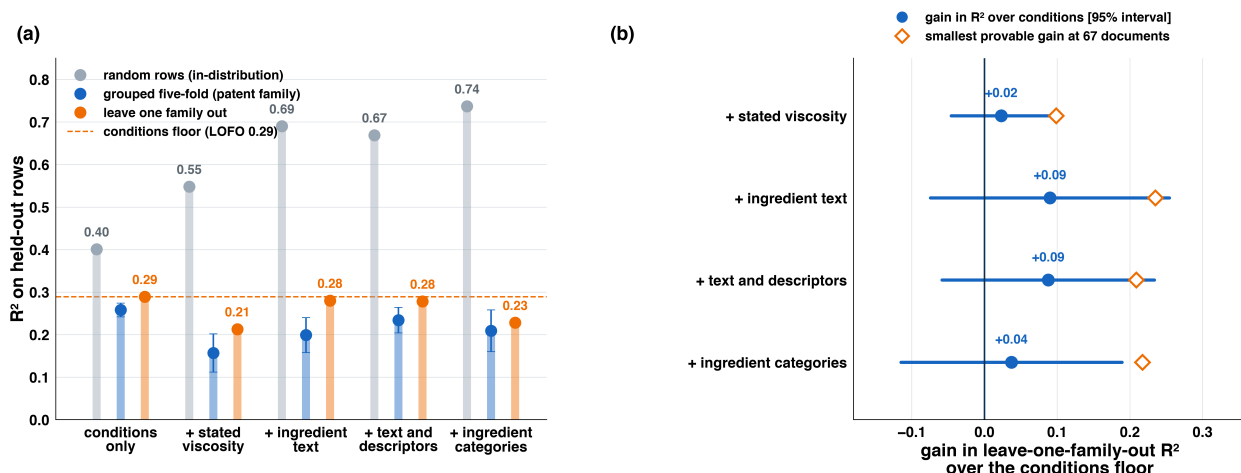


Figure 4: Chemistry arms do not separate from the conditions floor when patent families are held out. (a) R^2 for each rung of the ladder under random-row splitting (grey), grouped five-fold cross-validation with patent families held out (blue, bars are fold noise over 30 repeats), and leave-one-family-out evaluation (orange). The dashed line is the conditions floor. (b) Gain in leave-one-family-out R^2 over the same-estimator conditions floor with its 95% interval from leaving out one document family at a time (blue), and the smallest gain a paired test could distinguish from zero at 67 documents (open diamond).

(panel a), the base polymers, crosslinkers, catalysts, initiators, inhibitors, and fillers form separate regions, so the encoder does read the chemistry the strings carry rather than their surface form. Panel (b) shows the 233 base-polymer descriptions alone, colored by the reactive group the taxonomy pass assigned. Vinyl-functional polydimethylsiloxanes (153 strings) fill most of the map and separate by how they are described, from bare “vinyl-terminated linear dimethylpolysiloxane” to grades with a stated viscosity and vinyl content. Fluorosilicones form their own island. Descriptions that name no reactive unit (7) and those whose reactivity is not stated (42) gather along one edge with generic strings such as “silicone rubber base”. Silanol-terminated polymers for condensation cure (30) are not separated from vinyl polymers of similar description, which is a limit of a general-text encoder that the taxonomy labels are meant to compensate. The annotated strings are the medoids of eight clusters in embedding space, in the form the encoder receives them, which is the form a reader of the release sees as well. Distances in these maps are qualitative and do not establish predictive generalization.

6 Lessons from verification

A handful of cases changed the pipeline’s design and are worth recording, because each is a failure mode that a reader of the release should know was looked for.

A correct value can be attached to the wrong chemistry. Two reprints of one patent family (an EP and a US member) agreed on 27 of 27 compression-set values and on 0 of 9 ingredient sets, because the two extractions had read different tables as the composition. Values reproduce across duplicate extractions; recipes do not. That finding is why every composition in the release traces to a rendered page and why the duplicate-family census is part of every freeze.

A model can invent a plausible chemical name. One extraction returned “vinyltrimethylsiloxy-

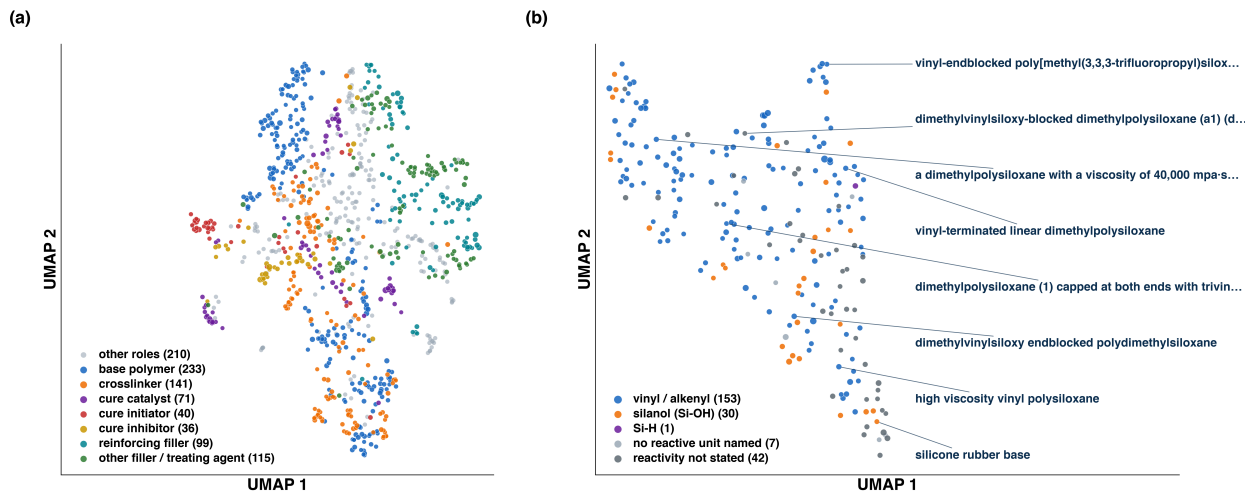


Figure 5: Ingredient descriptions cluster by role and, within the base polymers, by reactive group. UMAP projection of MatSciBERT embeddings of the 945 distinct ingredient descriptions in release cs_v14. (a) All descriptions colored by taxonomy role, with marker area proportional to the number of formulations using the string. (b) The 233 base-polymer descriptions alone, colored by the reactive group the taxonomy pass assigned; annotations are the medoid description of each of eight embedding-space clusters, trimmed to the chemistry clause.

terminated” where the source read “vinyl dimethylsiloxy-terminated”. The invented name is a valid substituent pattern and would pass a chemistry-only check; only comparison to the source catches it. Source checking and chemical plausibility are complementary and neither replaces the other.

Machine reconstruction of complex tables was, at one stage, the dominant source of defective compositions. An adversarial audit found that the admission rule for reconstructed recipes had a precision of 26.9%, and that 190 of the 644 rows in the first pinned cohort rested on such reconstructions. Those rows were re-transcribed from page images or quarantined, and the current release admits no reconstruction that has not been checked against the page.

Identical compositions with different measured values are a detector, not a ceiling. In a census of such collisions 54% were parser failures (a dropped column, a merged row) and the rest were dropped variables such as a post-cure or a different specimen. The collisions now route to a repair loop.

The document that most enriches the aging data carries no chemistry term in its title. A seal-test survey with 63 compression-set rows would have been lost to a title-only filter; it is the reason screening reads full text.

7 Companion benchmark: tables from a 1968 reference volume

The table-reading tools were developed in parallel on a setting where every answer can be checked. Noll’s 1968 monograph on organosilicon chemistry contains hundreds of pages of tabulated boiling points at stated pressures for named compounds, many drawn as structures. It is scanned, some tables are rotated, footnotes fall outside table crops, and the same compound appears under several names, so it exercises every failure the patent corpus shows while offering a fixed ground truth. A gold set was built by two blind readers plus a parser, with reconciliation only on low-confidence disagreements.

Two extraction arms were compared on matched pages. A fixed pipeline (page screening, orientation detection, table region detection, vision transcription, name-to-structure resolution) recovered 412 of 514 gold structures exactly and 350 of 585 complete boiling-point and pressure pairs. A tool-assisted arm, in which the model may zoom, rotate, separate a name from a formula, or re-query a parser after a refusal, recovered 430 of 514 structures and 388 of 585 pairs on the same parser. The verified release holds 509 measurements over 436 structures with 159 references, 73 of them with a DOI. A Gaussian-process model on a 786-row modeling merge predicts boiling point with a random-split test MAE of 15.7 °C and about 19 °C under compound-grouped and scaffold-grouped splits, with calibrated uncertainty. The grouping units there are molecules and Murcko scaffolds, which are the molecular analogues of the patent family used in the silicone work; the protocol is the same, the unit differs. The tools and the two-reader gold protocol are the parts of that work that transfer to patents, and a matched pilot on patent pages is the first item in Section 8.

8 Next steps and requests to collaborators

Three experiments are planned. The first is a matched pilot of the tool-assisted reader on patent pages already in the release, scored on the same audit sample as Section 4.3, with correct complete records per review hour as the metric. The second is adjudication of the candidate polymer labels (end group, reactive-unit placement, backbone monomers) that a whole-document pass has produced for 326 base polymers in 66 of the 67 release documents; 76 need review, and the labels enter the feature set only after that review. The third is the next freeze, `cs_v15`, which will carry the audit’s confirmed corrections and any new documents that clear the screen.

Two inputs from collaborators would change what can be done. A small set of formulations with chemistry and test conditions that Los Alamos trusts, with or without measured values, would serve as a fixed validation set on which representation choices can be compared without the noise of transfer to unseen documents. And a statement of the prediction error that would change a design decision, in percentage points of compression set at a named protocol, would let the corpus size needed to prove a useful model be estimated from the smallest-provable-gain curve rather than assumed. Which silicone families (addition-cured liquid rubbers, peroxide-cured high consistency rubbers, condensation-cured RTVs, foams) and which test regimes matter most would set the priority for the next acquisition round.

A Data dictionary of the release tables

The clean release consists of three tables joined on `formulation_id` and `doc_id` (Table 4). Two provenance tables repeat every column and add the fields in Table 5.

B Reproducibility

Every number in this report is either read from the release manifest or regenerated by `technical_report/make_figures.py`, which writes the five figures and the macro file `generated/numbers.tex` that the text includes. The script reads only committed artifacts. The release snapshot is `3_machine_learning_model/data/cs_v14/` (manifest commit `1b15dfa4`, database build 2026-09-13T19:18:58Z); the ladder metrics are `3_machine_learning_model/results/cs_v14/summary.csv`; the corpus-funnel stages after the downloaded-PDF starting count and the row-versus-document

Table 4: Clean release tables.

formulations.csv (one ingredient of one formulation)	
<code>formulation_id</code>	Release-local key (<code>doc_id</code> plus running number).
<code>doc_id</code>	Source document; joins <code>documents.csv</code> .
<code>ingredient</code>	Published description of the ingredient, chemistry first, then stated characterization. This is the string the text-embedding arm reads.
<code>role</code>	Primary role from the reviewed taxonomy; <code>unknown_role</code> when the pass abstained.
<code>amount</code>	Amount as printed.
<code>amount_basis</code>	<code>parts_by_weight</code> , <code>wt_percent_final_formulation</code> , <code>wt_pct</code> , <code>mass_g</code> , <code>ppm</code> , or <code>unspecified</code> .
compression_set.csv (one measurement)	
<code>measurement_id</code>	Release-local key.
<code>formulation_id</code>	Specimen’s formulation; joins <code>formulations.csv</code> .
<code>compression_set_pct</code>	Compression set in percent (ASTM D395 / ISO 815 definition).
<code>temperature_C</code> , <code>time_h</code> , <code>deflection_pct</code>	Test protocol; present on every released row.
<code>medium</code>	air, humid, water, or the oil named; empty when not stated.
<code>specimen_shape</code> , <code>specimen_thickness_mm</code> , <code>specimen_diameter_mm</code>	Geometry where stated.
<code>test_standard</code>	One spelling per standard (ASTM D395, ISO 815-1, JIS K 6262, JIS K 6249); empty when not named.
documents.csv (one document)	
<code>doc_id</code> , <code>source_type</code> , <code>title</code> , <code>year</code> , <code>doi</code> , <code>url</code>	Identity and location of the source.
<code>n_formulations</code> , <code>n_measurements</code>	Released rows from this document.

condition coverage are computed from the production DuckDB store of the same build. The audit sample, briefs, and both readers’ verdicts are under `rtv_foam_spr/data/gold/table_fidelity_audit/cs_v14/` and are never read by the pipeline.

The release chain is a sequence of job submissions. The database is rebuilt and the release frozen in `rtv_foam_spr` (dataset stages 10a and 10b), exported as a static snapshot in `3_machine_learning_model` (stage 00), and the ladder is run one shard per estimator and arm (stage 10) followed by the report stage (11). The figures in this report are produced by `technical_report/scripts/qsub/make_figures.sh`. All language-model passes run on locally served open-weight models with the model identifier and prompt hash recorded in every cache row; no document text leaves the institution’s cluster.

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Table 5: Provenance columns added in `formulations_full.csv` and `compression_set_full.csv`.

<code>name_verbatim</code> , <code>name_canonical</code> , <code>resolution_method</code>	The printed ingredient string, the resolved name, and the resolver branch that produced it.
<code>name_quality</code> , <code>name_quality_reason</code>	Judged quality of the published string (specific, generic_typed, opaque) with a one-sentence reason.
<code>sample_label</code> , <code>page</code> , <code>property_verbatim</code> , <code>measurement_conditions</code>	The row as printed and the page it was read from.
<code>cond_source</code> , <code>condition_assumed</code>	Where temperature, time, and deflection came from (human_verified, doc_stated_test_condition, llm, regex, manual_assumption) and whether the protocol was assumed.
<code>is_aging</code> , <code>post_cure_flag</code> , <code>post_cure_temp_c</code> , <code>post_cure_time_h</code>	Pre-aging and post-cure where the source states them.
<code>cv_group</code>	Patent-family-aware evaluation group.
<code>released</code>	Whether the row appears in the clean table.

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